

Nassau Candy Distributor

Shipping Route Efficiency Analysis

Data-Driven Logistics Optimization

Prepared for: Nassau Candy Distributor | Unified Mentor Internship Project
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Tools Used: Python · Pandas · Plotly · Streamlit

10,194 Total Orders	1,320.8 days Avg Lead Time	33.1% Delay Rate	20 Routes Analyzed
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1. Introduction

Abstract

This report presents a comprehensive data-driven analysis of shipping route efficiency for Nassau Candy Distributor, a national candy distributor operating across four US regions. Using order and shipment data containing 10,194 transactions from 2024 to 2030, we calculated key performance indicators including shipping lead time, delay rates, and route efficiency scores. The analysis reveals that the company's average shipping lead time is 1,320 days (approximately 3.6 years), indicating severe operational bottlenecks across all factory-to-customer routes. We identified the most efficient route as Sugar Shack -> Gulf with 1,091 days, while the least efficient route is Sugar Shack -> Pacific with 1,516 days. Based on these findings, we provide three actionable recommendations to reduce lead times, optimize shipping methods, and resolve geographic bottlenecks.

1.1 Problem Statement

Nassau Candy Distributor operates as a national distributor shipping products from multiple factories to customers across four US regions: Atlantic, Gulf, Interior, and Pacific. Despite having rich order and shipment data, the organization currently lacks clarity on:

- Which factory-to-customer routes are consistently efficient
- Which routes experience frequent delays
- How shipping performance varies by region, state, and ship mode
- Where operational bottlenecks exist geographically

Without this visibility, logistics optimization remains reactive rather than data-driven. Shipping inefficiencies directly affect customer satisfaction, increase operational costs, and reduce scalability. This project transforms raw order and shipment data into route-level operational intelligence, enabling data-driven logistics decisions.

1.2 Project Objectives

- Calculate and benchmark shipping lead time as the primary KPI
- Identify top 10 most efficient and bottom 10 least efficient routes
- Analyze regional and state-level performance to detect bottlenecks
- Compare shipping methods to find optimal cost-speed tradeoffs
- Build an interactive Streamlit dashboard for ongoing logistics monitoring

2. Dataset Description

2.1 Data Overview

The dataset contains 10,194 orders placed between January 2024 and December 2025, with shipments scheduled from June 2026 to June 2030. Each record represents a single order line item with complete customer, product, and shipping information across 19 fields.

2.2 Fields Description

Field	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date when order was placed (DD-MM-YYYY)
Ship Date	Date when order was shipped (DD-MM-YYYY)
Ship Mode	Shipping method (Standard, Second, First Class, Same Day)
Customer ID	Unique customer identifier
Country/Region	Country of customer (USA only)
City	City of customer
State/Province	State or province of customer
Postal Code	Zip code of customer
Division	Product division (Chocolate, Sugar, Other)
Region	Customer region (Atlantic, Gulf, Interior, Pacific)
Product ID	Unique product identifier
Product Name	Full product name
Sales	Total sales value of order (\$)
Units	Total units ordered
Gross Profit	Profit = Sales - Cost
Cost	Manufacturing cost (\$)

2.3 Factory Mapping

Each product is manufactured at one of five factories. Product-to-factory mapping was performed using a Python dictionary lookup, adding a Factory column to the dataset which serves as the origin point for all route definitions.

Factory	State	Coordinates	Products Manufactured
Lot's O' Nuts	Arizona	32.88, -111.77	Wonka Bar - Nutty Crunch Surprise, Fudge Mallows
Wicked Choccy's	Georgia	32.08, -81.09	Wonka Bar - Milk Chocolate, Triple Dazzle Caramel
Sugar Shack	Minnesota	48.12, -96.18	Laffy Taffy, SweeTARTS, Nerds, Fun Dip
Secret Factory	Illinois	41.45, -90.57	Everlasting Gobstopper, Lickable Wallpaper

The Other Factory	Tennessee	35.12, -89.97	Hair Toffee, Kazookles
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3. Methodology

3.1 Data Cleaning

Step 1 - Date Parsing: Dates were in DD-MM-YYYY format and parsed explicitly.

Step 2 - Lead Time Calculation: The primary KPI - shipping lead time - was derived as:

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df["Lead Time"] = (df["Ship Date"] - df["Order Date"]).dt.days
```

Step 3 - Invalid Row Removal: Rows with negative lead time (ship date before order date) were removed.

3.2 Feature Engineering

Route Definition: Each unique factory-to-customer region combination was defined as a route.

Delay Flag: A shipment was flagged as delayed if lead time exceeded mean + 1 standard deviation.

4. Results and Analysis

4.1 Critical Finding - Severe Operational Delays

Analysis reveals that Nassau Candy's average shipping lead time is 1,320 days (approximately 3.6 years), indicating severe operational bottlenecks. Even the most efficient route (Sugar Shack -> Gulf) takes nearly 3 years (1,091 days).

4.2 Route Efficiency Analysis

Top 5 Most Efficient Routes

Rank	Route	Avg Lead Time (days)	Efficiency Score	Total Orders
1	Sugar Shack -> Gulf	1,091.2	100.0	42
2	The Other Factory -> Interior	1,206.6	72.9	113
3	The Other Factory -> Gulf	1,272.6	57.4	194
4	The Other Factory -> Atlantic	1,282.7	55.0	38
5	Secret Factory -> Interior	1,289.1	53.5	45

Bottom 5 Least Efficient Routes

Rank	Route	Avg Lead Time (days)	Delay Rate	Total Orders
1 (worst)	Sugar Shack -> Pacific	1,516.7	66.7%	3
2	Sugar Shack -> Atlantic	1,375.4	44.4%	18
3	Secret Factory -> Atlantic	1,349.2	41.7%	72
4	Secret Factory -> Gulf	1,332.5	43.2%	37
5	Wicked Choccy's -> Pacific	1,329.1	35.3%	1,342

4.3 Regional Performance

Region	Orders	Avg Lead Time (days)	Delay Rate	Total Sales
Atlantic	2,986	1,322.8	32%	\$41,197
Gulf	1,620	1,311.4	32%	\$22,247
Interior	2,335	1,323.1	34%	\$32,038
Pacific	3,253	1,322.2	34%	\$46,302

5. Conclusion and Recommendations

5.1 Summary of Findings

- Critical Lead Time Issue: Average shipping lead time of 1,320 days (3.6 years)
- Route Performance Variance: 425-day difference between fastest and slowest routes
- Ship Mode Inefficiency: Premium shipping methods perform worse than Standard Class
- Geographic Bottlenecks: North Dakota and West Virginia have 100% delay rates

5.2 Three Key Recommendations

Recommendation 1: Investigate Data Accuracy vs. Operational Reality

- Confirm whether ship dates reflect actual shipment dates or future planned dates
- If actual: launch an urgent operational audit of warehouse processes
- If planned: redesign data capture to record actual ship dates

Recommendation 2: Standardize and Optimize Shipping Operations

- Eliminate or restructure "Same Day" and "First Class" shipping offerings
- Implement route-specific performance targets
- Create incentive programs for logistics partners

Recommendation 3: Address Geographic Bottlenecks

- Investigate North Dakota and West Virginia (100% delay rate)
- Consider regional distribution centers in high-delay states
- Implement real-time tracking for bottleneck states

Appendix: Dashboard Features

The accompanying Streamlit dashboard provides an interactive interface for ongoing logistics monitoring. It is structured into four tabs:

Tab	Features
Route Efficiency	Leaderboard with efficiency color scale
Geographic View	US choropleth map with state-level lead time
Ship Mode Analysis	Cost vs speed comparison charts
Route Drill-Down	Per-route lead time distribution